Call for Case Study Proposals — Pilot Research Software Engineering Course (Spring 2026, University of Delaware)

We invite industry, government, and national laboratory partners to submit case studies for the **Pilot Research Software Engineering (RSE) Course (CISC 867010)** at the University of Delaware, led by **Prof. Sunita Chandrasekaran**.

This graduate-level course trains students from diverse disciplines to apply **software engineering, AI/ML, and high-performance computing (HPC)** principles to real-world research challenges.

Selected case studies will be developed by **teams of 3–4 graduate students** under faculty mentorship. These projects are designed to strengthen university–industry collaboration while giving students hands-on experience in applied research software development.

Submission deadline: Jan 15, 2026 (Classes begin on Feb 02, so this will need to be a hard deadline)

Contact: Prof. Sunita Chandrasekaran (schandra@udel.edu)

🔷 Section 1 — Partner Information

1. Organization / Company Name *

Best Egg

2. Department / Group / Division

Technology Operations

3. Primary Contact Name *

Mike Urban

4. Title / Role *

Chief Techology Operations Officer

5. Contact Email *

mike@bestegg.com

6. Contact Phone (optional)

7. Organization Type *

- (●) Industry
- ( ) Government Lab
- ( ) Nonprofit
- ( ) Higher Education
- ( ) Other: ____________

🔷 Section 2 — Case Study Overview

8. Proposed Project Title *

SENTINEL (Scalable, ENabled, Trustworthy Infrastructure for Next-genAI Execution Layer)

9. Problem Statement — What do you expect the student team to do? *
(Provide 2–4 sentences on the technical challenge or research question.)

How do take the next step in our journey with GenAI in integrating more unstructured data to that will help the business make better decisions and enable even broader AI adoption and value add and do that in a scalable, resilient, and secure way.

Help define and create a scalable, resilient and secure genAI infrastructure and orchestration layer.

10. Impact — Why does this problem matter to your organization? *
(Describe the relevance, background, or impact.)

We have delivered a number of high value work productivity solutions leverage just Enterprise ChatGPT and have a couple closer to the customer custom AI pilot solutions in the works. A unlock for us is leveraging more unstructured data and integrating with more of our solutions. This deeper integration brings potential more business risk and being in the financial services Industry where we are highly regulated we have to deliver in a scalable, resilient, and secure way.

11. Expected Outcomes or Deliverables *
(e.g., prototype code, workflow, visualization dashboard, ML model, report, etc.)

Framework, visualization dashboard of how this framework Is delivering on scalability, resiliency, and security. I think It would also be great If the students are able to build the Infrastructure that allows this to happen.

12. Desired Skills or Background for Student Team *
(e.g. python, machine learning, data engineering, data analysis, software engineering, prompt engineering, fine tuning LLM)

software engineering, prompt engineering, Infrastructure engineering, observability engineering and cyber security engineering

🔷 Section 3 — Collaboration & Logistics

13. Estimated Compute Needs *

- ○ Small-scale (laptop or cloud compute)
- ◉ Moderate (university cluster)
- ○ Large-scale (HPC/GPU resources)
- ○ Other:

14. We need a point person (in-person or virtual) from your team for the students to contact during the course of the semester. Who will be this point person? *

(Name and email)

Mike Urban - mike@bestegg.com (May have other contacts for folks on my various teams to help as well)

15. To ensure the success of both the course and its participants, the designated point person from your organization is expected to attend ~ 4 key checkpoints scheduled during class hours.

- [x] We acknowledge

16. Would your organization be open to continued collaboration beyond this course, as there may be an opportunity to extend the project into the summer?

- ◉ Yes
- ○ No
- ○ Maybe

Section 4 — Optional Uploads

17. Supporting Materials (optional)
(Attach any short project summary, diagram, or publication reference)

Add file

Section 5 — Consent & Acknowledgment

18. Confirmation *

- [x] I understand that this form is to express interest in submitting a case study for the Pilot RSE Course at the University of Delaware. Faculty will review submissions and contact selected partners for next steps.